US 20250273892A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0273892 A1
(43) Pub. Date: Aug. 28, 2025
SHIH et al.

(54) **MODULE MOUNTING DEVICE**

(71) Applicant: **Fulian Precision Electronics (Tianjin) Co., LTD**, Tianjin (CN)

(72) Inventors: **TUNG-HO SHIH**, New Taipei (TW); **HSUAN-AN SHIH**, New Taipei (TW)

(21) Appl. No.: **18/931,298**

(22) Filed: **Oct. 30, 2024**

(30) **Foreign Application Priority Data**

Feb. 27, 2024 (CN) 202410215141.9

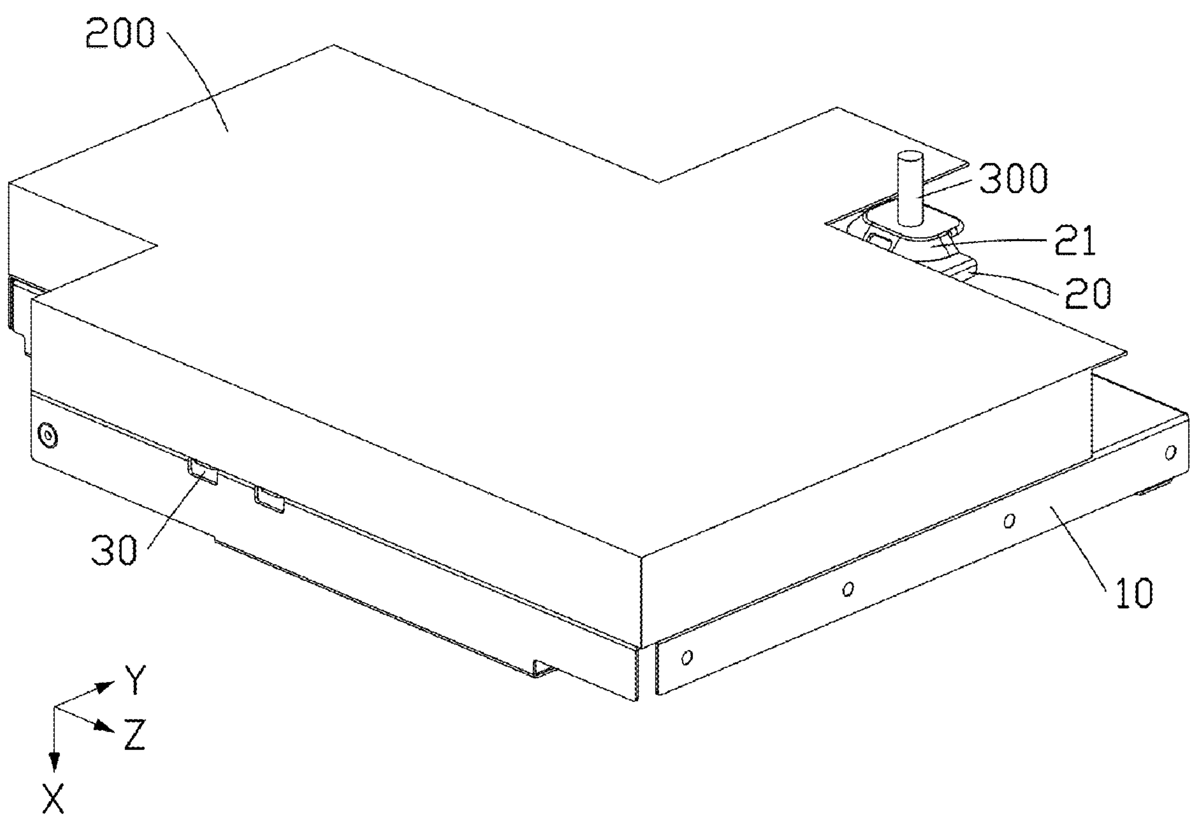
Publication Classification

(51) **Int. Cl.**
H01R 13/24 (2006.01)
F16B 5/06 (2006.01)

(52) **U.S. Cl.**
CPC ***H01R 13/2421*** (2013.01); ***F16B 5/0628*** (2013.01); ***F16B 5/0657*** (2013.01)

(57) **ABSTRACT**

A module mounting device includes a bracket, a mounting member, a connecting rod, and a probe assembly. The bracket can mount a first module. The mounting member is slidably connected to the bracket, the mounting member includes a mounting portion, the mounting portion can mount a second module. The connecting rod is rotatably connected to the bracket and is slidably connected to the mounting member. The probe assembly is disposed on the bracket, a portion of the probe assembly extends through the bracket and is connected to the connecting rod. Wherein the probe assembly is configured to drive the connecting rod to rotate under a pressing force from the first module, and the connecting rod is configured to drive the mounting portion to protrude relative to the bracket.



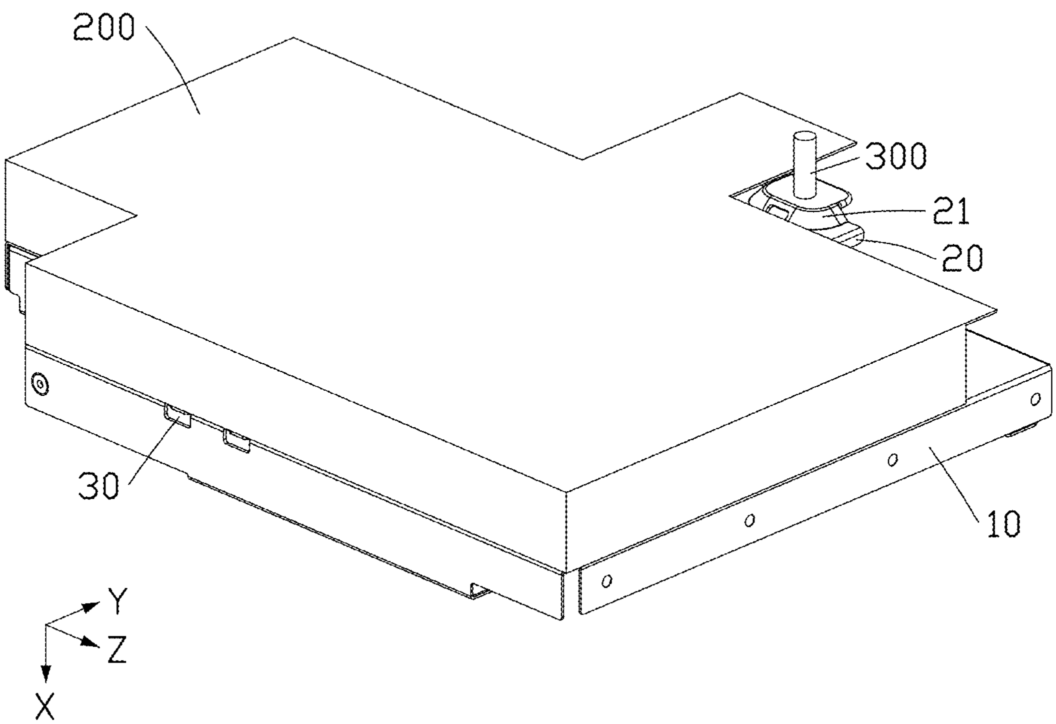


FIG. 1

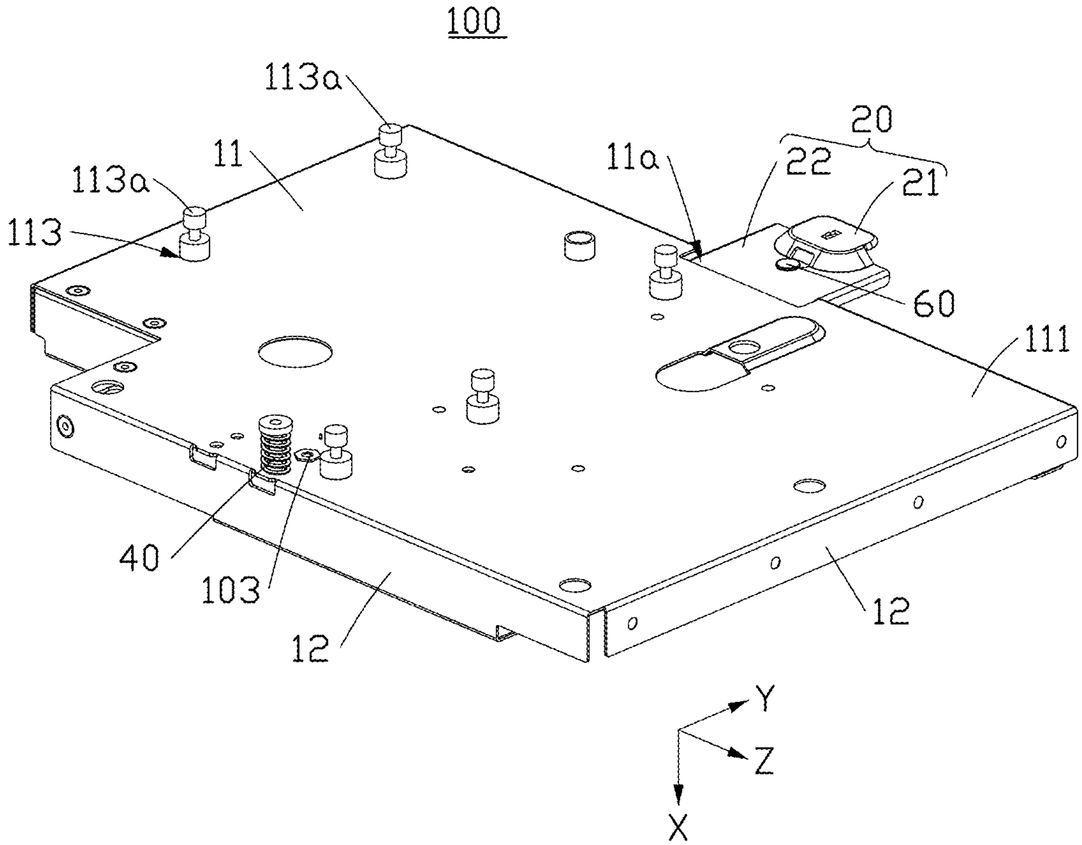


FIG. 2

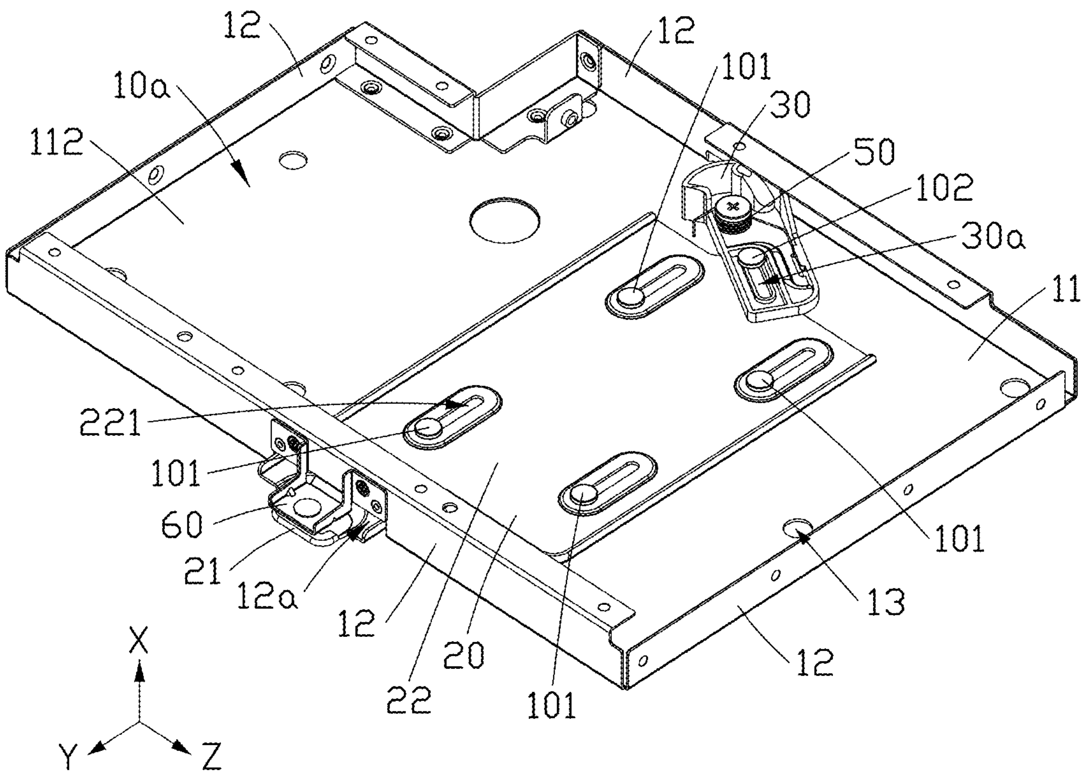

12
12
10a
101
30
112
50
102
30a
11
221
101
101
60
21
12a
13
12
12
20
22
101
X
Y
Z

FIG. 3

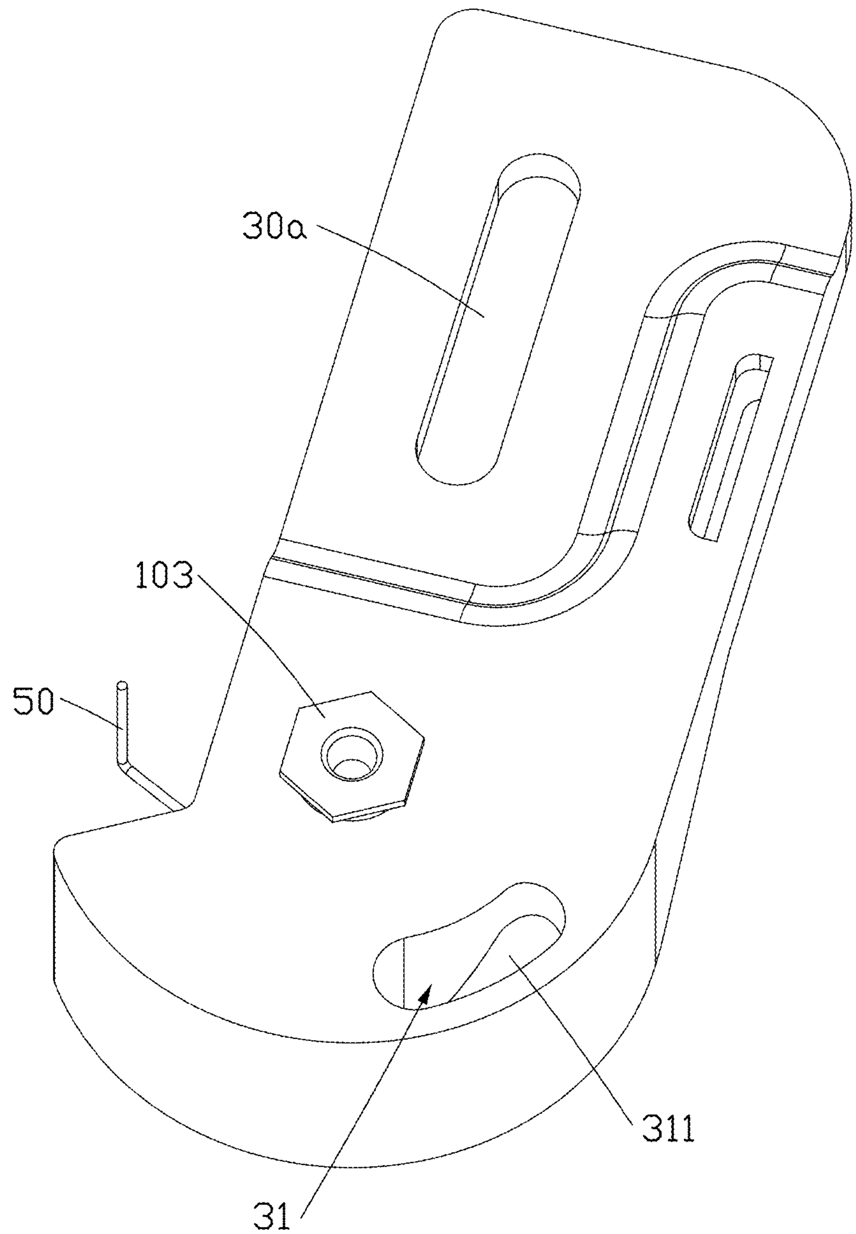


FIG. 4

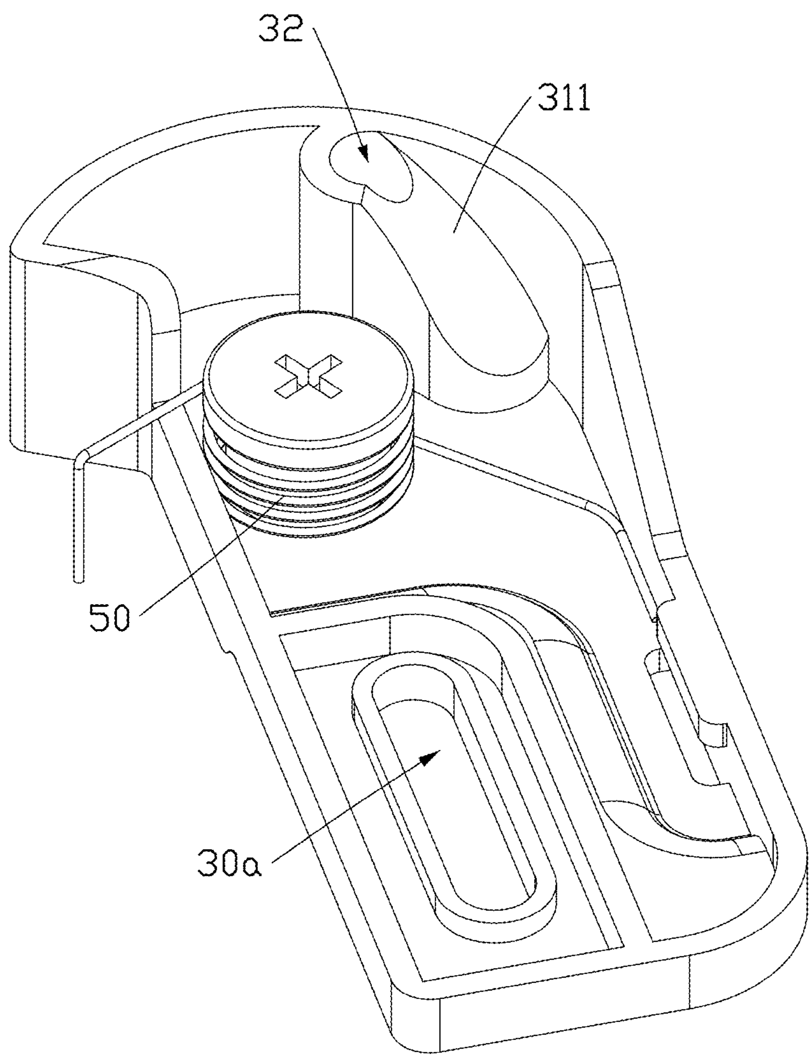


FIG. 5

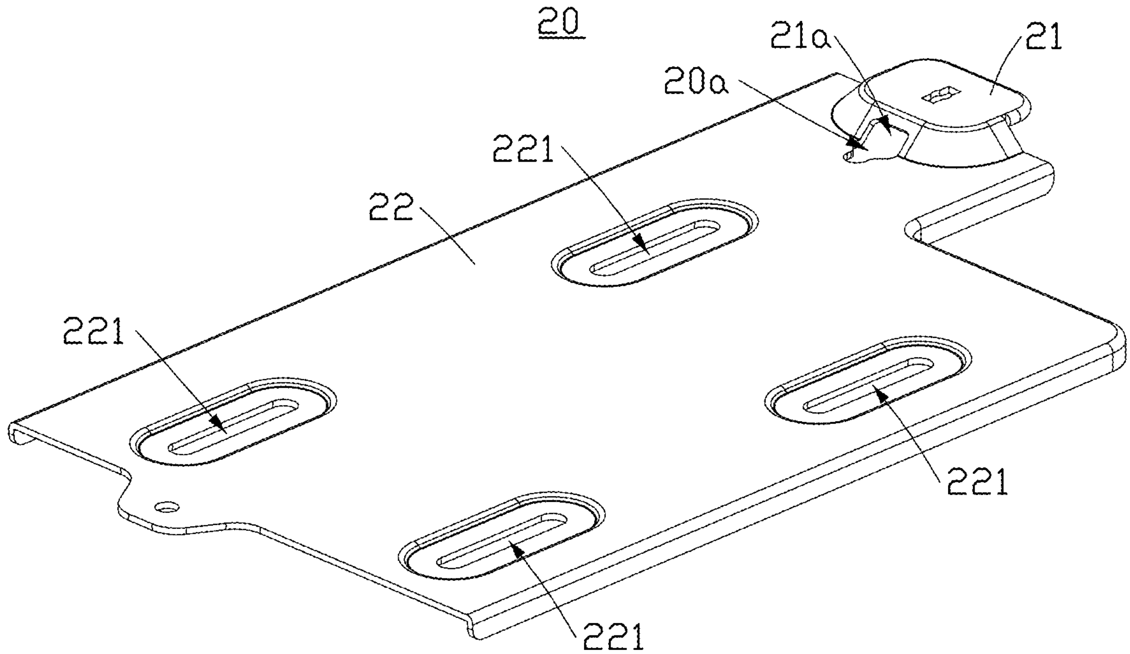


FIG. 6

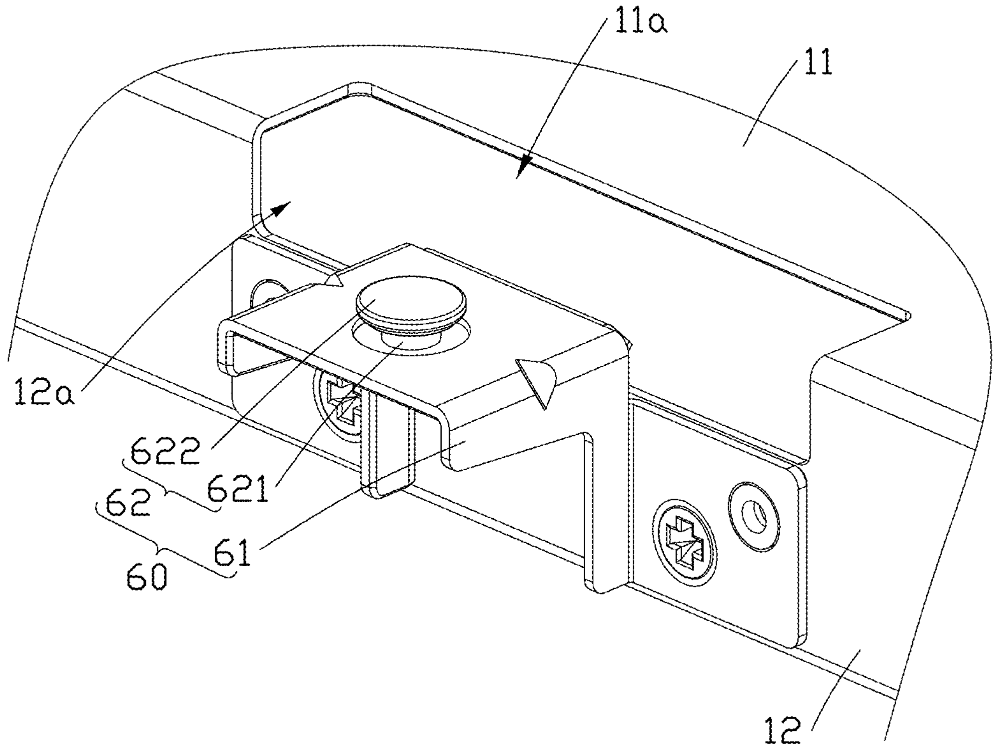


FIG. 7

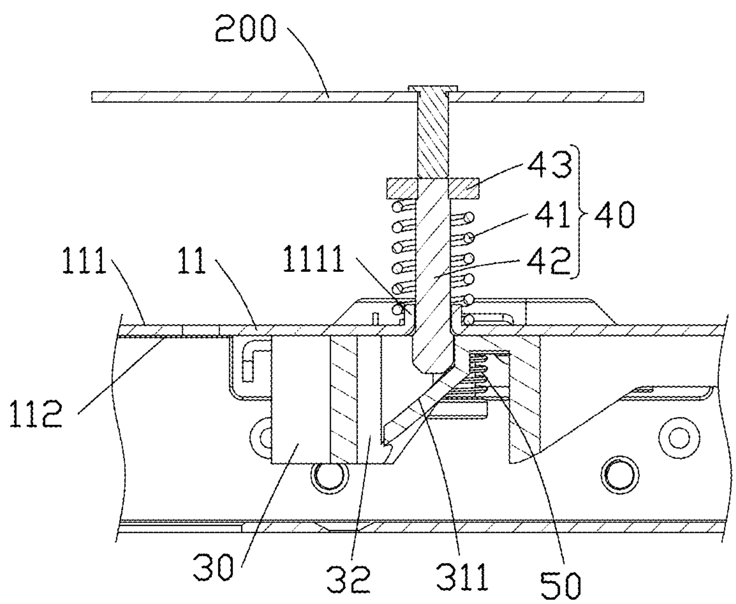


FIG. 8

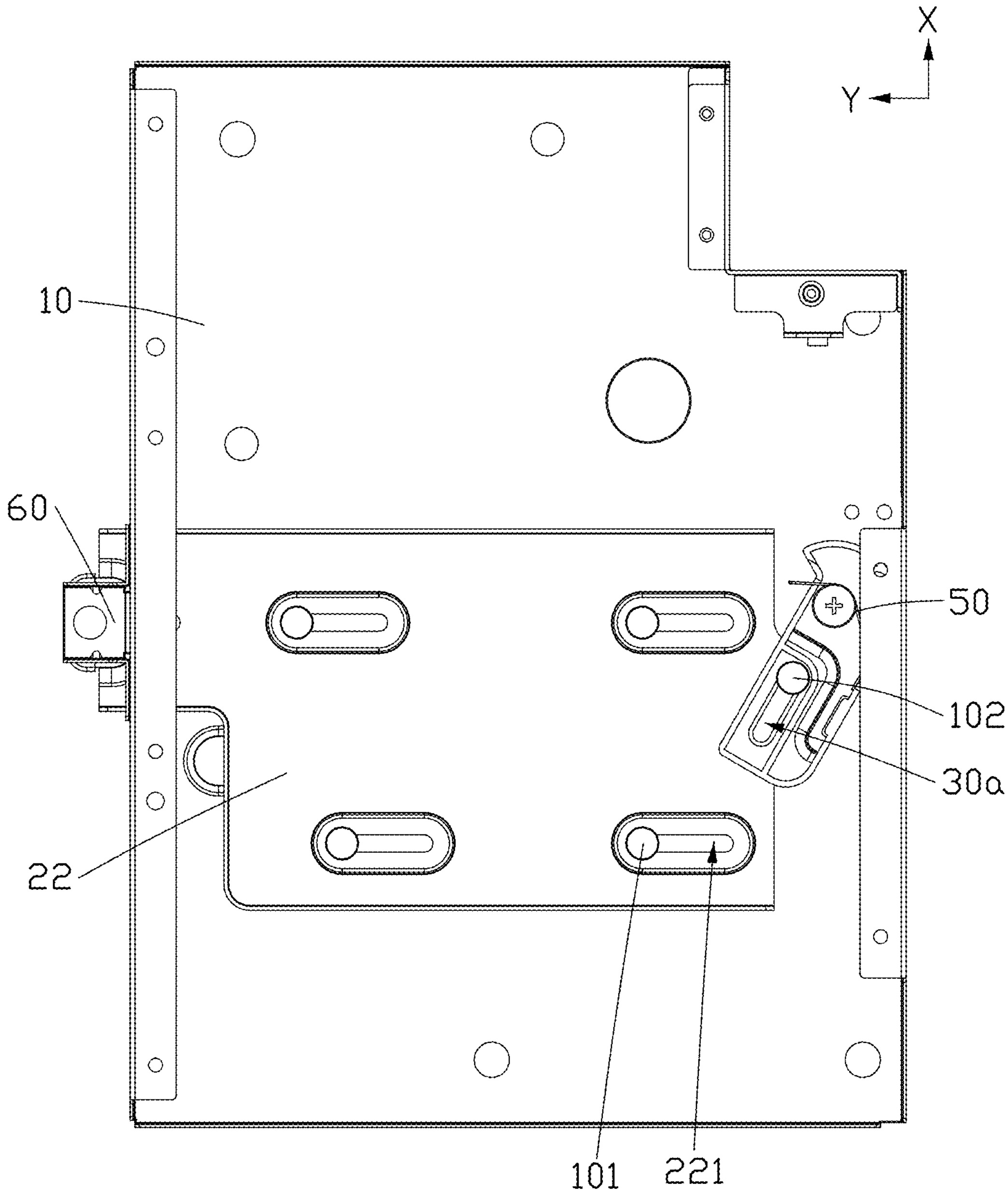


FIG. 9

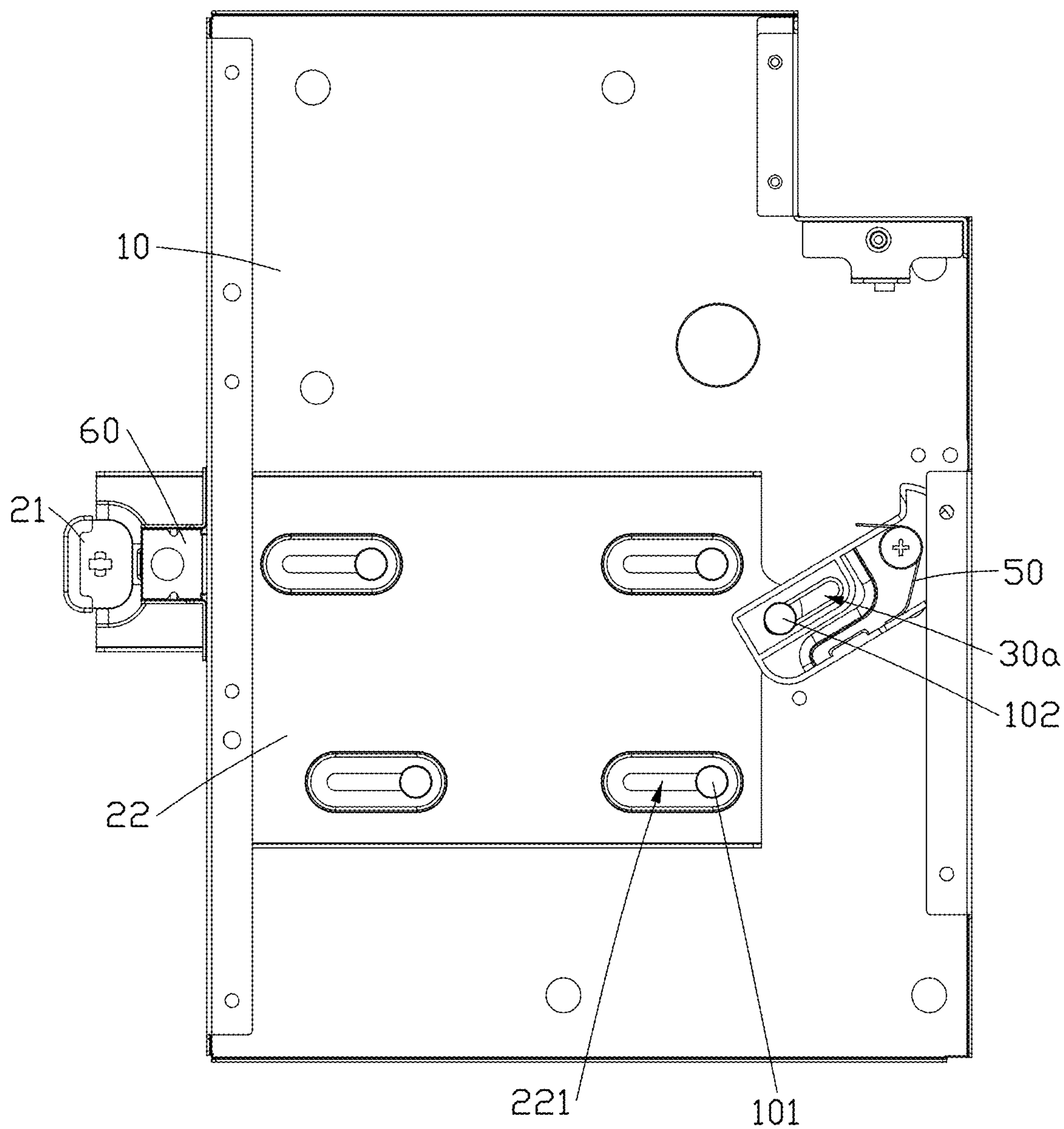


FIG. 10

MODULE MOUNTING DEVICE

FIELD

[0001] The subject matter herein generally relates to fixtures, and more particularly, to a module mounting device.

BACKGROUND

[0002] When different modules are assembled into a same space, due to limitation of the space, there is a risk of interference between the different modules during assembling, which increases a difficulty of the assembly process and reduces assembly efficiency. Therefore, there is a room for improvement in the art.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Implementations of the present technology will now be described, by way of example only, with reference to the attached figures.

[0004] FIG. **1** is a diagrammatic view including a module mounting device, a first module, and a second module according to an embodiment of the present disclosure.

[0005] FIG. **2** is a diagrammatic view of the module mounting device of FIG. **1**.

[0006] FIG. **3** is similar to the FIG. **2**, but showing the module mounting device from another angle.

[0007] FIG. **4** is a diagrammatic view of a connecting rod of the module mounting device in FIG. **3**.

[0008] FIG. **5** is similar to the FIG. **4**, but showing the connecting rod from another angle.

[0009] FIG. **6** is a diagrammatic view of a mounting member of the module mounting device in FIG. **2**.

[0010] FIG. **7** is a diagrammatic view of a limiting member of the module mounting device in FIG. **2**.

[0011] FIG. **8** is a cross-sectional view of a portion of the module mounting device and the first module of FIG. **1**.

[0012] FIG. **9** is a diagrammatic view showing the mounting member of FIG. **2** in a retracted state.

[0013] FIG. **10** is a diagrammatic view showing the mounting member o of FIG. **2** in a protruding state.

DETAILED DESCRIPTION

[0014] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale, and the proportions of certain parts may be exaggerated to better illustrate details and features of the present disclosure.

[0015] The term "comprising," when utilized, means "including, but not necessarily limited to"; it specifically indicates open-ended inclusion or membership in the so-described combination, group, series, and the like.

[0016] Some embodiments of the present disclosure will be described in detail with reference to the drawings. If no conflict, the following embodiments and features in the embodiments can be combined with each other.

[0017] Referring to FIGS. **1-3**, a module mounting device **100** is provided according to an embodiment of the present disclosure. The module mounting device **100** includes a bracket **10**, a mounting member **20**, a connecting rod **30**, and a probe assembly **40**. The bracket **10** includes a top wall **11**. The top wall **11** includes a first mounting surface **111** and a second mounting surface **112** opposite to each other in a first direction X. A first module **200** can be mounted on the first mounting surface **111**. The mounting member **20** is slidably connected to the second mounting surface **112**. The mounting member **20** includes a mounting portion **21** for mounting a second module **300**. The connecting rod **30** is rotatably connected to the second mounting surface **112** and slidably connected to the mounting member **20**. The probe assembly **40** is disposed on the first mounting surface **111**. A portion of the probe assembly **40** extends through the first mounting surface **111** and is connected to the connecting rod **30**. When the first module **200** is installed, the probe assembly **40** can be pushed by the first module **200**, and the probe assembly **40** drives the connecting rod **30** to rotate, which in turn drives the mounting portion **21** to slide and extend relative to the bracket **10**.

[0018] When installing the first module **200**, the first module **200** drives the probe assembly **40** to press the connecting rod **30**, and the connecting rod **30** rotates to drive the mounting portion **21** to protrude relative to the bracket **10**, which facilitates the installation of the second module **300**. When the mounting portion **21** does not protrude relative to the bracket **10**, an interference between the mounting portion **21** and other modules may be reduced, which reduces the assembly difficulty and improves assembly efficiency.

[0019] Referring to FIGS. **2**, **3**, and **9**, in some embodiments, the top wall **11** is provided with a plurality of connecting holes **113**. Each connecting hole **113** extends through the top wall **11**. The connecting hole **113** is provided with a protruding portion **113***a* therein for connecting the first module **200**, so that the first module **200** can be mounted on the first mounting surface **111**.

[0020] In some embodiments, the bracket **10** further includes a sidewall **12** connected to the top wall **11**. The top wall **11** and the sidewall **12** cooperatively define a receiving space **10***a*. The connecting rod **30** and a portion of the mounting member **20** are disposed in the receiving space **10***a*, thereby reducing a space occupied by the connecting rod **30** and the mounting member **20**, improving space utilization and reducing an overall space occupied by the module mounting device **100**.

[0021] In some embodiments, the top wall **11** is further provided with a first opening **11***a*, and the sidewall **12** is provided with a second opening **12***a*. The first opening **11***a* is communicated with the second opening **12***a*, thereby facilitating the installation of the mounting member **20** into the bracket **10**. The mounting portion **21** protrudes from the first opening **11***a* and the second opening **12***a*.

[0022] Referring to FIGS. **3**, **4** and **6**, in some embodiments, the mounting member **20** includes a main body **22** connected to the mounting portion **21**. The mounting portion **21** can protrude from the main body **22** in a direction opposite to the first direction X.

[0023] In some embodiments, the main body **22** is provided with at least one first chute **221** extending along a second direction Y. The second direction Y is perpendicular to the first direction X. The main body **22** is slidably connected to the second mounting surface **112** through the first chute **221**.

[0024] In some embodiments, the main body **22** is provided with a plurality of first chutes **221**. The main body **22** is provided with four first chutes **221**, and the four first sliding slots **221** are disposed in a rectangular shape.

[0025] In some embodiments, the module mounting device **100** further includes a first fixing member **101**. The first fixing member **101** passes through the first chute **221** and is fixed to the top wall **11**, so that the mounting member **20** can slide relative to the top wall **11** along the second direction Y.

[0026] In some embodiments, the connecting rod **30** is provided with a second chute **30***a*, and the main body **22** is slidably connected to the second chute **30***a*. The module mounting device **100** further includes a second fixing member **102**. The second fixing member **102** passes through the second chute **30***a* and is fixed to the top wall **11**, so that the connecting rod **30** is slidably connected to the main body **22**. An extending direction of the second chute **30***a* intersects with both the first direction X and the second direction Y.

[0027] In some embodiments, a connection area between the second fixing member **102** and the top wall **11** is disposed between adjacent first chutes **221** in a third direction Z, which is beneficial for the main body **22** to slide relative to the top wall **11**. The third direction Z is perpendicular to the first direction X and the second direction Y.

[0028] Referring to FIGS. **2**, **3**, and **5** to **7**, in some embodiments, the module mounting device **100** further includes a connecting shaft **103** fixed to the top wall **11**. The connecting rod **30** is rotatably connected to the connecting shaft **103**. The connecting rod **30** can rotate relative to the connecting shaft **103**.

[0029] In some embodiments, a recess **31** is defined on the connecting rod **30**. The recess **31** includes an inclined surface **311**. The probe assembly **40** is connected to the inclined surface **311**. When the probe assembly **40** is pressed, the connecting rod **30** is driven to rotate relative to the connecting shaft **103**, and the main body **22** is driven to slide along the second sliding groove **30***a* and the first sliding groove **221**. The mounting portion **21** protrudes relative to the bracket **10** and the first module **200**.

[0030] In some embodiments, a positioning hole **32** is defined on the connecting rod **30**. The positioning hole **32** extends through the recess **31** and connects to a lower end of the inclined surface **311**. After the first module **200** is mounted on the first mounting surface **111**, a portion of the probe assembly **40** is disposed in the positioning hole **32**, thereby fixing the mounting portion **21** and the connecting rod **30**.

[0031] In some embodiments, the probe assembly **40** includes a first elastic member **41**, a probe **42**, and a connecting member **43**. One end of the first elastic member **41** is connected to the first mounting surface **111**, and another end of the first elastic member **41** is connected to the connecting member **43**. The probe **42** is connected to the connecting member **43**, and the probe **42** extends through the first elastic member **41** and is connected to the inclined surface **311**.

[0032] In some embodiments, the first mounting surface **111** is provided with a protrusion **1111**. One end of the first elastic member **41** is sleeved on the protrusion **1111**. The first elastic member **41** can be a spring, the first elastic member **41** can elastically deformable in along the first direction X.

[0033] In some embodiments, the module mounting device **100** further includes a second elastic member **50** connected to the connecting rod **30**. The second elastic member **50** drives the connecting rod **30** to rotate, thereby causing the connecting rod **30** to drive the mounting member **20** to retract. The second elastic member **50** can be a torsion spring, and the second elastic member **50** can generate torsional deformation around the first direction X.

[0034] When the first module **200** is mounted on the first mounting surface **111**, the first elastic member **41** is compressed. The probe **42** drives the connecting rod **30** to rotate, and the connecting rod **30** will apply a force during the rotation onto the second elastic member **50**. When the first module **200** is disassembled, the first elastic member **41** rebounds, thereby driving the probe **42** to disengage from the positioning hole **32**, and the second elastic member **50** drives the connecting rod **30** to rotate in an opposite direction, causing the connecting rod **30** to drive the mounting member **20** to retract.

[0035] Referring to FIGS. **2** to **4** and **8** to **10**, in some embodiments, the module mounting device **100** further includes a limiting member **60**. The limiting member **60** includes a limiting body **61** and a limiting portion **62**. The limiting body **61** is connected to the bracket **10**, and the limiting portion **62** is connected to the limiting body **61**. The mounting member **20** is provided with a limiting hole **20***a*. When the mounting portion **21** protrudes from the main body **22**, the limiting portion **62** is accommodated in the limiting hole **20***a*.

[0036] In some embodiments, the limiting body **61** is fixed to the sidewall **12**.

[0037] In some embodiments, the mounting portion **21** is provided with a through hole **21***a*. The limiting hole **20***a* extends through the main body **22**, and the through hole **21***a* communicates with the limiting hole **20***a*. When the mounting portion **21** protrudes from the main body **22**, the limiting portion **62** extends through the through hole **21***a* and is clamped in the limiting hole **20***a*, which facilitates the positioning of the mounting portion **21**.

[0038] In some embodiments, the limiting portion **62** includes a first limiting area **621** and a second limiting area **622**. The first limiting area **621** is connected to the limiting body **61**, and the second limiting area **622** is connected to the first limiting area **621**. When the first module **200** is mounted on the first mounting surface **111**, the first limiting area **621** is clamped in the limiting hole **20***a*, and the second limiting area **622** covers at least a portion of the limiting hole **20***a*, which can enhance the stability of the mounting member **20** and prevent the first limiting area **621** from disengaging from the limiting hole **20***a*.

[0039] Referring to FIGS. **1**, **9**, and **10**, when the module mounting device **100** is in use, the first module **200** is first installed. When the first module **200** is mounted on the first mounting surface **111**, the first module **200** compresses the first elastic member **41**, and the connecting member drives the probe **42** to press against the inclined surface **311** and be accommodated in the positioning hole **32**. The probe **42** then drives the connecting rod **30** to rotate, thereby causing the connecting rod **30** to drive the mounting member **20** to slide, allowing the mounting portion **21** to protrude relative to the bracket **10**. The first limiting area **621** is then clamped in the limiting hole **20***a*, and the second module **300** is installed on the mounting portion **21**. When the first module **200** is disassembled, the first elastic member **41** rebounds and drives the probe **42** to protrude from the positioning hole **32**, and the second elastic member **50** drives the connecting rod **30** to rotate in the opposite direction, which in turn drives the mounting member **20** to retract. The second module **300** can be disassembled before or after disassembling the first module **200**.

[0040] When installing the first module **200**, the first module **200** drives the probe assembly **40** to press the connecting rod **30**, and the connecting rod **30** rotates to drive the mounting portion **21** to protrude relative to the bracket **10**, which facilitates the installation of the second module **300**. When the mounting portion **21** does not protrude relative to the bracket **10**, an interference between the mounting portion **21** and other modules may be reduced, which reduces the assembly difficulty and improves assembly efficiency. The mounting member **20**, connecting rod **30**, and probe assembly **40** can all automatically reset, which reduces operational time and enhances disassembly efficiency.

[0041] It is to be understood, even though information and advantages of the present embodiments have been set forth in the foregoing description, together with details of the structures and functions of the present embodiments, the disclosure is illustrative only; changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the present embodiments to the full extent indicated by the plain meaning of the terms in which the appended claims are expressed.

What is claimed is:

1. A module mounting device comprising:

a bracket comprising a top wall, the top wall comprising a first mounting surface and a second mounting surface opposite to each other in a first direction, the first mounting surface configured to mount a first module;

a mounting member slidably connected to the second mounting surface, the mounting member comprising a mounting portion, the mounting portion configured to mount a second module;

a connecting rod rotatably connected to the second mounting surface and slidably connected to the mounting member; and

a probe assembly disposed on the first mounting surface, a portion of the probe assembly extending through the first mounting surface and connected to the connecting rod;

wherein the probe assembly is configured to drive the connecting rod to rotate under a pressing force from the first module, and the connecting rod is configured to drive the mounting portion to protrude relative to the bracket.

2. The module mounting device of claim **1**, wherein the connecting rod defines a recess, the recess comprises an inclined surface, and the probe assembly is connected to the inclined surface.

3. The module mounting device of claim **2**, wherein the connecting rod further defines a positioning hole, the positioning hole extends through the recess, and a portion of the probe assembly is disposed in the positioning hole when the first module is mounted on the first mounting surface.

4. The module mounting device of claim **2**, wherein the probe assembly comprises a first elastic member, a probe, and a connecting member, one end of the first elastic member is connected to the first mounting surface, another end of the first elastic member is connected to the connecting member, the probe is connected to the connecting member, the probe extends through the first elastic member and is connected to the inclined surface.

5. The module mounting device of claim **4**, wherein the first elastic member is a spring, and the first elastic member is elastically deformable in the first direction.

6. The module mounting device of claim **1**, further comprising a limiting member, wherein the limiting member comprises a limiting body and a limiting portion, the limiting body is connected to the bracket, and the limiting portion is connected to the limiting body, the mounting member defines a limiting hole, and the limiting portion is accommodated in the limiting hole when the mounting portion protrudes relative to the bracket.

7. The module mounting device of claim **6**, wherein the mounting member further comprises a main body connected to the mounting portion, the mounting portion is configured to protrude relative to the bracket in a direction opposite to the first direction, the mounting portion defines a through hole, and the limiting hole extends through the main body and communicates with the limiting hole.

8. The module mounting device of claim **6**, wherein the limiting portion comprises a first limiting area and a second limiting area, the first limiting area is connected to the limiting body, and the second limiting area is connected to the first limiting area, the first limiting area is clamped in the limiting hole when the first module is mounted on the first mounting surface, and the second limiting area covers at least a portion of the limiting hole.

9. The module mounting device of claim **6**, wherein the bracket further comprises a sidewall connected to the top wall, the top wall and the sidewall cooperatively define a receiving space, and the connecting rod is disposed in the receiving space.

10. The module mounting device of claim **9**, wherein the top wall further defines a first opening, the sidewall defines a second opening communicating with the second opening, and the mounting portion protrudes from the first opening and the second opening.

11. The module mounting device of claim **7**, wherein the main body further defines at least one first chute, the main body is slidably connected to the second mounting surface through the at least one first chute, the connecting rod defines a second chute, and the main body is slidably connected to the second chute.

12. The module mounting device of claim **11**, wherein the at least one first chute extending along a second direction, and an extending direction of the second chute intersects with both the first direction and the second direction.

13. The module mounting device of claim **11**, wherein the second direction is perpendicular to the first direction.

14. The module mounting device of claim **1**, further comprising a second elastic member, wherein the second elastic member is connected to the connecting rod, the second elastic member is configured to drive the connecting rod to rotate, thereby causing the connecting rod to drive the mounting member to retract.

15. The module mounting device of claim **14**, wherein the second elastic member is a torsion spring, and the second elastic member is configured to generate torsional deformation around the first direction.

* * * * *